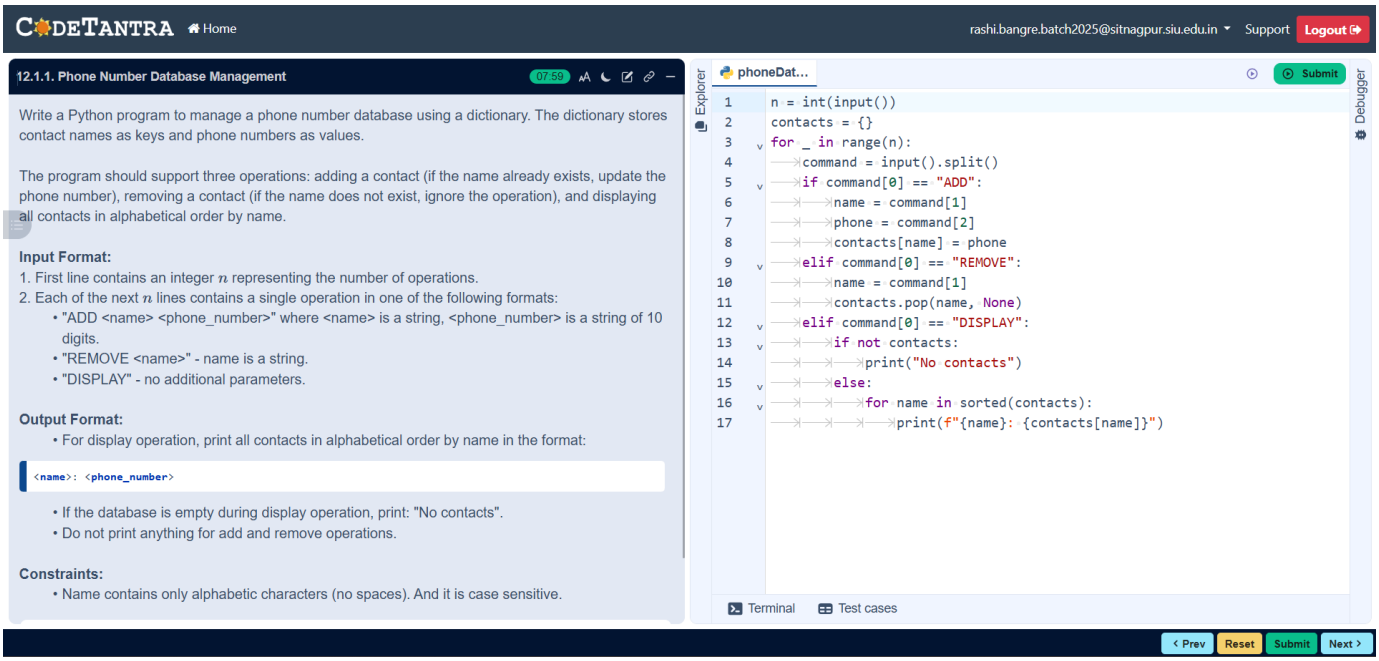
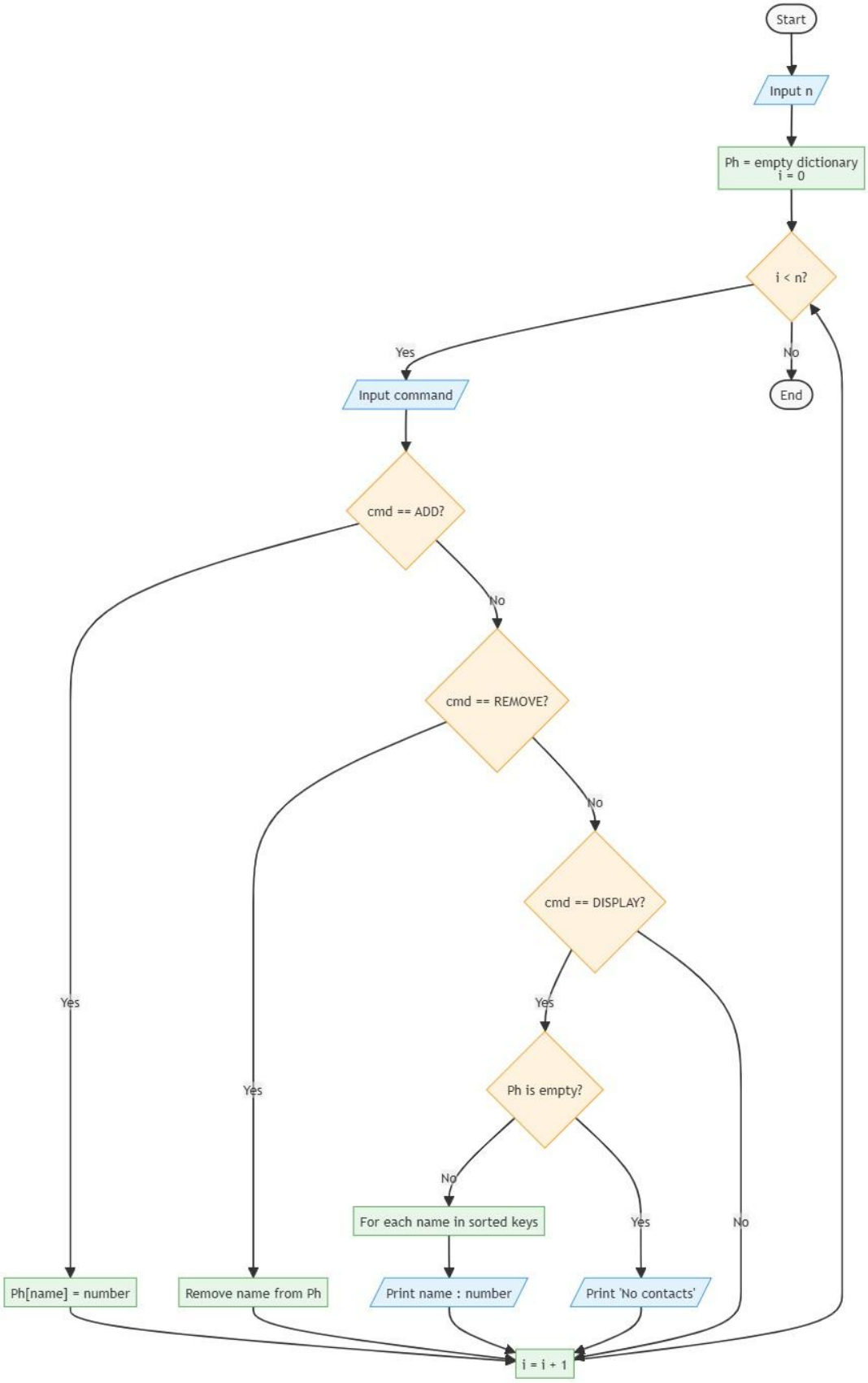
12.1.1 Phone Number Database Management

ALGORITHM

1. START
2. READ an integer n representing the total number of operations to perform.
3. INITIALIZE an empty dictionary named Ph.
4. FOR i from 1 to n:
   - READ the operation command and parameters as a list cmd.
   - IF cmd[0] is "ADD":
     - Add or update the contact: Ph[cmd[1]] = cmd[2].
   - ELSE IF cmd[0] is "REMOVE":
     - Remove the contact if it exists using Ph.pop(cmd[1], None).
   - ELSE IF cmd[0] is "DISPLAY":
     - IF the dictionary Ph is empty:
       - PRINT "No contacts".

       ELSE:
       - SORT the dictionary keys (names) alphabetically.
       - FOR each name in the sorted keys:
       - PRINT the contact in the format {name}: {phone_number}.
5. END




Start
Input n
Ph = empty dictionary
i = 0
i < n?
Yes
No
End
Input command
cmd == ADD?
No
cmd == REMOVE?
No
cmd == DISPLAY?
Yes
Yes
Yes
Ph is empty?
No
Yes
No
For each name in sorted keys
Ph[name] = number
Remove name from Ph
Print name : number
Print 'No contacts'
i = i + 1